

# OLTP VS OLAP

# OLTP (ON-LINE TRANSACTION PROCESSING)

- is characterized by a large number of short on-line transactions (INSERT, UPDATE, DELETE).
- The main emphasis for OLTP systems is put on very fast query processing, maintaining data integrity in multi-access environments and an effectiveness measured by number of transactions per second.
- In OLTP database there is detailed and current data, and schema used to store transactional databases is the entity model (usually 3NF).



# OLAP (ON-LINE ANALYTICAL PROCESSING)

- is characterized by relatively low volume of transactions.
- Queries are often very complex and involve aggregations.
- For OLAP systems a response time is an effectiveness measure.
- OLAP applications are widely used by Data Mining techniques.
- In OLAP database there is aggregated, historical data, stored in multi-dimensional schemas (usually star schema).



# OLTP vs. OLAP

- OLTP: On-Line Transaction Processing

- Many short transactions (queries + updates)
- Examples:
  - Update account balance
  - Enroll in course
  - Add book to shopping cart
- Queries touch small amounts of data (one record or a few records)
- Updates are frequent
- Concurrency is biggest performance concern

- OLAP: On-Line Analytical Processing

- Long transactions, complex queries
- Examples:
  - Report total sales for each department in each month
  - Identify top-selling books
  - Count classes with fewer than 10 students
- Queries touch large amounts of data
- Updates are infrequent
- Individual queries can require lots of resources



# Why OLAP & OLTP don't mix (1)

## Different performance requirements

- Transaction processing (OLTP):
  - Fast response time important (< 1 second)
  - Data must be up-to-date, consistent at all times
- Data analysis (OLAP):
  - Queries can consume lots of resources
  - Can saturate CPUs and disk bandwidth
  - Operating on static “snapshot” of data usually OK
- OLAP can “crowd out” OLTP transactions
  - Transactions are slow → unhappy users
- Example:
  - Analysis query asks for sum of all sales
  - Acquires lock on sales table for consistency
  - New sales transaction is blocked



# Why OLAP & OLTP don't mix (2)

## Different data modeling requirements

### Transaction processing (OLTP):

- Normalized schema for consistency
- Complex data models, many tables
- Limited number of standardized queries and updates

### Data analysis (OLAP):

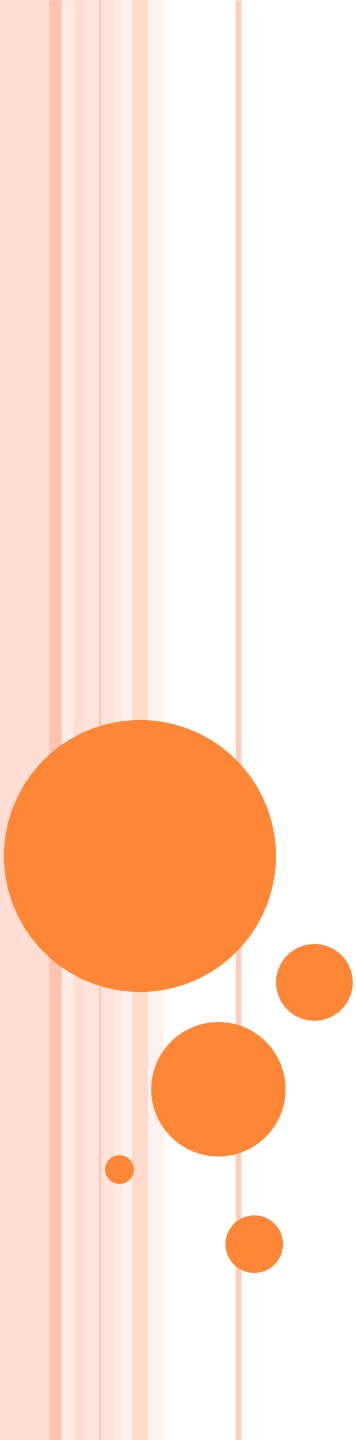
- Simplicity of data model is important
  - Allow semi-technical users to formulate ad hoc queries
- De-normalized schemas are common
  - Fewer joins → improved query performance
  - Fewer tables → schema is easier to understand



# Why OLAP & OLTP don't mix (3)

Analysis requires data from many sources

- An OLTP system targets one specific process
  - For example: ordering from an online store
- OLAP integrates data from different processes
  - Combine sales, inventory, and purchasing data
  - Analyze experiments conducted by different labs
- OLAP often makes use of historical data
  - Identify long-term patterns
  - Notice changes in behavior over time
- Terminology, schemas vary across data sources
  - Integrating data from disparate sources is a major challenge



# **DATA WAREHOUSING AND OPERATIONAL DBMS**



# WHAT IS DATA WAREHOUSING?

# DEFINITION:

- A data warehouse is a copy of transaction data specifically structured for querying and reporting.
- An expanded definition for data warehousing includes business intelligence tools, tools to extract, transform and load data into the repository, and tools to manage and retrieve metadata.



# NOTHING TO DO WITH WHETHER SOMETHING IS A DATA WAREHOUSE.

- This definition of the data warehouse focuses on data storage.
- A data warehouse can be normalized or denormalized.
- It can be a relational database, multidimensional database, flat file, hierarchical database, object database, etc.
- Data warehouse data often gets changed.
- And data warehouses often focus on a specific activity or entity.



# WHY DO WE NEED DATA WAREHOUSES?

- Consolidation of information resources
- Improved query performance
- Separate research and decision support functions from the operational systems
- Foundation for data mining, data visualization, advanced reporting and OLAP tools



- The data stored in the warehouse is uploaded from the operational systems.
- The data may pass through an operational data store for additional operations before it is used in the DW for reporting.
- Operational DBMS is used to deal with the everyday running of one aspect of an enterprise.
- OLTP (on-line transaction processor) or Operational DBMS are usually designed independently of each other and it is difficult for them to share information.



# HOW DO DATA WAREHOUSES DIFFER FROM OPERATIONAL DBMS?

- Goals
- Structure
- Size
- Performance optimization
- Technologies used



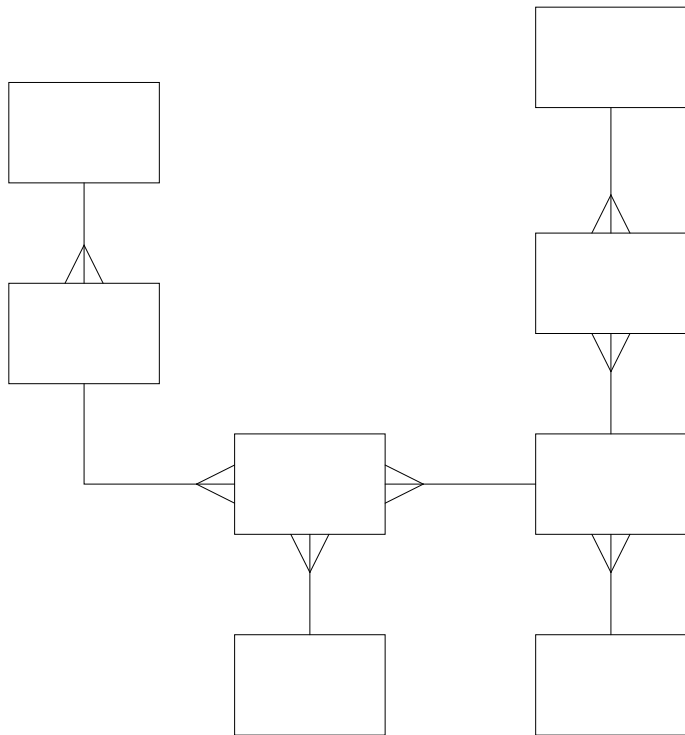
# HOW DO DATA WAREHOUSES DIFFER FROM OPERATIONAL SYSTEMS?

| <b>Data warehouse</b>                                                     | <b>Operational DBMS</b>                                                |
|---------------------------------------------------------------------------|------------------------------------------------------------------------|
| <b>Subject oriented</b>                                                   | <b>Transaction oriented</b>                                            |
| <b>Large (hundreds of GB up to several TB)</b>                            | <b>Small (MB up to several GB)</b>                                     |
| <b>Historic data</b>                                                      | <b>Current data</b>                                                    |
| <b>De-normalized table structure (few tables, many columns per table)</b> | <b>Normalized table structure (many tables, few columns per table)</b> |
| <b>Batch updates</b>                                                      | <b>Continuous updates</b>                                              |
| <b>Usually very complex queries</b>                                       | <b>Simple to complex queries</b>                                       |



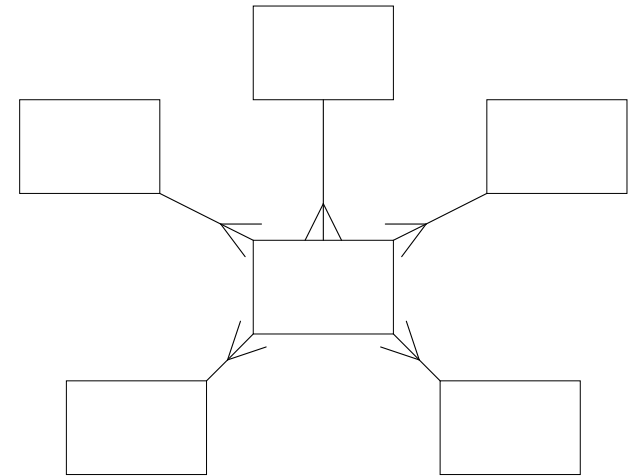
# DESIGN DIFFERENCES

## Operational DBMS



ER Diagram

## Data Warehouse



Star  
Schema



# FUNCTIONS

- A data warehouse maintains its functions in three layers: staging, integration, and access.
- Staging is used to store raw data for use by developers (analysis and support).
- The integration layer is used to integrate data and to have a level of abstraction from users.
- The access layer is for getting data out for users.



# WHAT IS A DATA WAREHOUSE USED FOR?

- Knowledge discovery
  - Making consolidated reports
  - Finding relationships and correlations
  - Data mining
  - Examples
    - Banks identifying credit risks
    - Insurance companies searching for fraud
    - Medical research

